



Figure 1: Artificial intelligence, machine learning and deep learning

most from AI. There are AI programs that can now predict the occurrence of cancer with 90 per cent accuracy just by analysing the symptoms of a patient, which can help a doctor to start treatment early.

Many of us often confuse the terms ‘artificial intelligence’, ‘machine learning’ and ‘deep learning’. Hence, we use the terms interchangeably. But these are not the same things.

Artificial intelligence, in broader terms, can be described as a branch of computer science that can imitate human beings. It has been demonstrated that computers can be programmed to carry out very complex tasks that were earlier only performed by humans. From self-driving cars and Amazon’s Alexa or Apple’s Siri, to a computer program playing a game of chess, all of these are applications of artificial intelligence.

Machine learning can be referred to as a subset of AI. It is considered one of the most successful approaches to AI, but is not the only approach. For example, there are many chat bots that are rule based, i.e., they can answer only certain questions, depending on the way they were programmed. But they will not be able to learn anything new from those questions. So this can be categorised as AI as the chat bots replicate human-like behaviour, but can’t be termed as machine learning. Now, the question is: can machines really ‘learn’? How is it possible for a machine to learn if it doesn’t have a brain and a complex nervous system like humans? According to Arthur Samuel, “Machine learning can be defined as a field of study that gives computers the ability to learn without being explicitly programmed.” We can also define it as the computer’s ability to learn from experience to perform a certain task, whereby the performance will improve with experience. This is akin to a computer program playing chess, which can be categorised as machine learning, if it learns from previous experiences and subsequently makes better moves to win a game.

Deep learning can be categorised as a subset of machine learning. It uses neural networks to simulate human decision-making skills. A neural network consists of many neurons and hence resembles a human nervous system. Have you ever wondered how Facebook detects your face amongst many, in an image? Image detection is one of the examples

of deep learning, which is much more complex as it needs lots of data to train itself. For instance, a deep learning algorithm can learn to recognise a car but will have to be trained on a huge data set which consists of cars as well as other objects. If this is not done, it might make a wrong decision like identifying a bus as a car. Hence, compared to other machine learning algorithms, a deep learning algorithm requires much more data in order to observe and understand every minute detail to make the right decisions.

From Figure 1, you can see how all these terms are related to each other, yet are not the same.

Now that you have understood the differences between artificial intelligence, machine learning and deep learning, let’s dig deeper into machine learning.

There are three main types of machine learning algorithms.

1. **Supervised learning:** The data set in supervised learning consists of input data as well as the expected output. The algorithm is a function which maps this input data to the expected outcome. The algorithm will continue to learn until the model achieves a desired level of accuracy. Then this model can be applied to new sets of data, for which the expected outcome is not available but needs to be predicted from a given set of data.

For instance, let’s look at a car manufacturing company that wants to set a price for its newest model. In order to do so, it can use this supervised learning model. The company’s input data set may consist of details of previous car models — their features like the number of air bags, electronic gadgets, etc. The output (or expected outcome) would be the sale price of the car. Now an algorithm can be designed to map those input parameters (or features) to the expected outcome.

Once the algorithm achieves the desired level of accuracy, this model can be applied to the firm’s new car model. This can help the company predict the car price at which it should be launched.

For better results, the company can use a data set of car models of other manufacturers and their prices. This would help the company in setting a competitive price.

In machine learning, the best results are not achieved by using a great algorithm but by using the most data.

2. **Unsupervised learning:** The only difference between supervised and unsupervised learning is that the data set doesn’t have the expected outcome as in the supervised learning model. The data set will only have input parameters (or features) and the algorithm will have to predict the outcome. For instance, if a shirt manufacturing company is looking to manufacture three different sizes of shirts (small, medium and large), its data comprises the shoulder, waist and chest sizes of its customers. Now, depending upon this huge data set, the company needs to group the sizes into three categories so that there can be a best fit for everyone. Here, an unsupervised learning model can be used to group the different data points in three different sizes and predict an appropriate shirt size for every customer.